Logistic Regression:

- Dataset: birthwt.csv
- Predictors: smoke, ht, ui
- Response variable: low

Regression Output:
Dataset: birthwt.csv
Rows used: 189

Logit Regression Results

Dep. Variable:	low	No. Observations:	189
Model:	Logit	Df Residuals:	185
Method:	MLE	Df Model:	3
Date:	Sun, 25 Oct 2026	Pseudo R-squ.:	0.06250
Time:	14:49:18	Log-Likelihood:	-110.00
converged:	True	LL-Null:	-117.34
Covariance Type:	nonrobust	LLR p-value:	0.002125

	coef	std err	z	P>|z|	[0.025	0.975]
const	-1.3551	0.241	-5.611	0.000	-1.828	-0.882
smoke	0.6831	0.329	2.075	0.038	0.038	1.329
ht	1.4174	0.624	2.273	0.023	0.195	2.640
ui	1.0384	0.428	2.426	0.015	0.200	1.877

Confusion Matrix:
[[119 11]
[45 14]]

Classification Report:

	precision	recall	f1-score	support
0	0.73	0.92	0.81	130
1	0.56	0.24	0.33	59
accuracy			0.70	189

```
   macro avg       0.64      0.58      0.57       189
weighted avg       0.67      0.70      0.66       189

Accuracy: 0.7037

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    accuracy                           0.70       189
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macro avg	0.64	0.58	0.57	189
weighted avg	0.67	0.70	0.66	189

Accuracy: 0.7037

1. The logistic regression model has an accuracy of 70.37%. This means that the model correctly classified about 70% of the observations in the dataset.
2. The lower precision is 0.56, or 56% for category 1 which represents low birthweight. This means that 56% of the cases predicted as low birthweight were actually low birthweight.
3. Yes both categories have at least 30 support. Category 0 has 130 observations, while category 1 has 59 observations.
4. The pseudo R^2 is appproximatley 0.0625, which is relatively low. This suggests that the model has limited explanatory power. Although its accuracy is around 70%, it does not perform very well at identifying low birthweight cases.

Count Variable Regression:

- Dataset: places_2025.csv
- Predictors: binge_crudeprev, checkup_crudeprev, csmoking_crudeprev, depression_crudeprev, obesity_crudeprev, sleep_crudeprev, selfcare_crudeprev, indeplive_crudeprev, loneliness_crudeprev, foodinsecu_crudeprev, housinsecu_crudeprev
- Response Variable: mhlth_crudeprev

Regression output:
Dataset: places_2025.csv
Rows used: 2299

Generalized Linear Model Regression Results

Dep. Variable:	mhlth_crudeprev	No. Observations:	2299
Model:	GLM	Df Residuals:	2287
Model Family:	Poisson	Df Model:	11
Link Function:	Log	Scale:	1.0000
Method:	IRLS	Log-Likelihood:	-5397.6
Date:	Sun, 25 Oct 2026	Deviance:	39.316
Time:	14:44:23	Pearson chi2:	39.2

No. Iterations: 4 Pseudo R-squ. (CS): 0.1925
Covariance Type: nonrobust

	coef	std err	z	P>|z|	[0.025	0.975]
const	1.7032	0.172	9.888	0.000	1.366	2.041
binge_crudeprev	0.0032	0.003	0.954	0.340	-0.003	0.010
checkup_crudeprev	-0.0004	0.002	-0.192	0.848	-0.004	0.004
csmoking_crudeprev	0.0043	0.004	1.100	0.271	-0.003	0.012
depression_crudeprev	0.0137	0.002	5.800	0.000	0.009	0.018
obesity_crudeprev	0.0001	0.002	0.076	0.940	-0.003	0.004
sleep_crudeprev	0.0024	0.002	1.059	0.289	-0.002	0.007
selfcare_crudeprev	-0.0608	0.022	-2.708	0.007	-0.105	-0.017
indeplive_crudeprev	0.0522	0.014	3.759	0.000	0.025	0.079
loneliness_crudeprev	0.0113	0.003	3.717	0.000	0.005	0.017
foodinsecu_crudeprev	-0.0083	0.007	-1.247	0.212	-0.021	0.005
housinsecu_crudeprev	0.0150	0.008	1.786	0.074	-0.001	0.032

McFadden Pseudo-R-squared: 0.043562

VIF Results:

Predictor	VIF
binge_crudeprev	2.528842
checkup_crudeprev	1.750444
csmoking_crudeprev	6.934794
depression_crudeprev	2.216376
obesity_crudeprev	2.573017
sleep_crudeprev	3.209334
selfcare_crudeprev	30.514413
indeplive_crudeprev	30.953309
loneliness_crudeprev	3.432565
foodinsecu_crudeprev	62.830787
housinsecu_crudeprev	46.839954

WARNING: The response is not count data.
Poisson regression assumptions are not met.

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foodinsecu_crudeprev	-0.0083	0.007	-1.247	0.212	-0.021	0.005
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McFadden Pseudo-R-squared: 0.043562

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1. The four significant predictors are depression_crudeprev, selfcare_crudeprev, indeplive_crudeprev, and lonliness_crudeprev. These predictors have p values below 0.05.
2. Based on the coefficient values selfcare_crudeprev has the largest absolute coefficient at approximately -0.0608. This means it has the strongest estimated effect per one unit increase among the four significant predictors, although coefficient size by itself does not determine statistical significance.
3. Yes the significant predictors selfcare_crudeprev and indeplive_crudeprev have VIF values of approximately 30.51 and 30.95. Both are above 10 which suggests high mulicollinearity.
4. The mcfadden pseudo R^2 is approximately 0.0436. This is a relatively low value, suggesting limited model fit. However pseudo R^2 does not directly represent the percentage of variabllitiy explained like regular R^2.